

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

Jyun-Hong Lin¹, Yang-Ming Yeh², Yu-Chun Chen¹, Yao-Ting Huang^{1,*}

¹Department of Computer Science and Information Engineering, National Chung Cheng University, Chiayi, Taiwan

²Department of Computer Science and Engineering, National Sun Yat-sen University, Kaohsiung, Taiwan

*Correspondence: ythuang@cs.ccu.edu.tw

Abstract

Long-read phasers rarely combine sequence variants with the methylation signal on the same reads, and seldom report which phase assignments are unreliable. Here we present LongPhase 2, which phases SNVs, indels, structural variants and 5-methylcytosine in one graph, weights observations by base quality and sequence context, and scores every variant's phasing uncertainty. A graph neural network identifies phase-disagreement onsets for selective unphasing. Scored against a benchmark from the telomere-to-telomere HG002 assembly, LongPhase 2 makes 2.8- to 4.8-fold fewer switch errors than WhatsHap and HapCUT2 on nanopore data while running five- to sixfold faster on one thread at 60×, and 2.2- to 2.6-fold fewer than WhatsHap on PacBio HiFi data without retraining. Co-phasing all four classes increases block N50 at 60× by 43% relative to SNV-only phasing. Against the earlier, mapping-based benchmark, the tools differ only 1.2- to 1.3-fold at high coverage, and most of their switch errors coincide. LongPhase 2 keeps the command-line interface and output formats of LongPhase, which Clair3, ClairS and EPI2ME call.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. Read-based phasing does not require parental sequencing, a population reference panel of the kind statistical phasing depends on⁵, or the additional Hi-C or Strand-seq libraries used to phase whole chromosomes⁶.

Read-based phasers optimize agreement between reads and candidate haplotypes through weighted minimum error correction or probabilistic models, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by re-aligning each read to both alleles, also co-phases small and large indels with them¹⁰. We previously introduced LongPhase¹¹, which recasts phasing as a weighted allele graph, resolves

haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{12,13}. WhatsHap and LongPhase are used in several long-read analysis pipelines (Supplementary Table 6). Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default^{14–17}; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase¹⁸; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes^{12,19}. Phasing errors can therefore propagate, unflagged, to downstream analyses.

Three challenges remain in read-based phasing. The first is integrating 5-methylcytosine (5mC), which long reads report alongside the sequence, into haplotype inference. Where the two haplotypes differ in methylation, as at imprinted loci and at sites of sequence-dependent allele-specific methylation, a read’s CpG states tie it to one haplotype even where it spans no heterozygous variant, which is where sequence-based blocks end^{20,21}. Several methylation-aware tools use that signal once a sequence phase exists. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved²⁰; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches^{21,22}; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves²³. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base^{20,21}; joint modelling allows such sparse, noisy evidence to be weighed together with the sequence alleles on the same reads.

The second challenge is accounting for errors that depend on genomic context. WhatsHap weights observations in its minimum-error-correction objective, and HapCUT2 incorporates base-quality information in its likelihood model^{7,8}. Base quality alone, however, cannot establish that a read originates from the locus to which it is aligned. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions can mislead the phasing graph^{24,25}. These context-dependent errors can create switches within haplotype blocks and confound compound-heterozygosity calls and haplotype-aware somatic detection^{16,19}.

A third challenge is identifying unreliable phase assignments. Statistical phasers attach posterior probabilities to phased genotypes⁵, and HapCUT2 uses posterior confidence to prune variants and split low-confidence blocks⁸. Such confidence estimates depend on the phasing model and may remain high when misplaced reads provide internally consistent evidence. Local sequence context and the surrounding evidence therefore provide additional information for assessing phase reliability. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read’s alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant’s phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning^{26,27}, also provide no per-variant reliability estimate. They learn, partly from simulated reads, to optimize the same fragment-cut or minimum-error-correction objective as heuristic phasers and phase SNVs only, and their authors note that errors from mis-alignment in repetitive regions remain unresolved²⁷.

Phasers are commonly benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38^{28,29}, which assess mainly the regions where mapping-based methods perform well³⁰. The HG002 v4.2.1 release omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications^{29,30}, where mis-mapped reads and collapsed paralogues create false linkages^{24,25}. Its phase is also partial: the original release recorded phase only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap²⁹, so the variants hardest for a read-based phaser are the least likely to be scored (the phase-transferred version scored here is described in Methods). Extending benchmarks into difficult sequence has changed conclusions before. GIAB’s growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set²⁹, and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample³¹. The complete diploid T2T-HG002 assembly extends the available phase information³⁰. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Its projection onto GRCh38, the GIAB v5.0q benchmark³², extends assessment into tandem repeats and homopolymers, although satellite arrays and many segmental duplications remain excluded.

Here we present LongPhase 2, which jointly phases sequence variation and allele-specific methylation, accounts for sequence context and identifies potentially unreliable phase assignments. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, letting methylation inform the phase while it is being built. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant, and a graph neural network, enabled by default, identifies phase-disagreement onsets for selective unphasing, leaving the genotypes unchanged. We evaluate it against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and it keeps the command-line interface and output formats of LongPhase.

Results

LongPhase 2 combines joint phasing with learned error detection

LongPhase 2 integrates SNVs, small indels, SVs and allele-specific 5mC sites called by its `modcall` module in a chromosome-wide phasing graph (Fig. 1a). Observations are weighted by base quality and sequence context, with filters for homopolymers, tandem repeats and candidate copy-number-altered regions (Fig. 1b). A graph neural network (GNN) examines windows centred on variants with high phasing uncertainty. The GNN predicts phase-disagreement onsets. Variants at these predicted onsets are then selectively unphased, their genotypes are retained, and blocks disconnected by unphasing are split (Fig. 1c). We call this step GNN correction. The outputs are phased VCFs for each variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 reads at 10–60× and PacBio HiFi reads from the same individual at 10–50× (Methods). Nanopore data comprised ten down-sampling replicates per coverage at 10–20× in 2× steps, and one each at 30, 40, 50 and 60×; HiFi and unphased-variant analyses used one replicate per coverage. All phasers received the



Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, Each variant receives a phasing-uncertainty score (Methods), and a graph neural network (GATv2 and Transformer layers; Supplementary Fig. 6) classifies the variants around the most uncertain ones. Predicted phase-disagreement onsets are selectively unphased (grey); we call this step GNN correction. **d**, Output: phased haplotypes of all four variant classes, with the selected variant left unphased (faded), and haplotagging reads.

same alignments and variant calls. We scored phase against two HG002 truth sets: GIAB v4.2.1, integrated from mapping-based call sets, and GIAB v5.0q, phased by the diploid telomere-to-telomere (T2T) assembly. Both truth sets were used in full, including variants outside their benchmark regions. Unless stated otherwise, results refer to SNV phasing against v5.0q and include GNN correction. Per-run values and summaries are in Supplementary Data 1 and Supplementary Tables 7–17.

LongPhase 2 reduces switch error rates across evidence types and sequencing platforms

In SNV-only phasing, LongPhase 2 had lower switch error rates than WhatsHap and HapCUT2 at all ten coverages, decreasing from 0.082% at 10 $\times$ to 0.023% at 60 $\times$ (Fig. 2a). The separation widened up to 20 $\times$ and then remained stable. Rates were 2.8- and 3.3-fold lower than those of WhatsHap and HapCUT2 at 10 $\times$, respectively, and 3.4–3.7- and 4.3–4.8-fold lower at 20–60 $\times$ (Supplementary Table 8). This difference held in every paired replicate at 10–20 $\times$ (2.8–4.1-fold versus WhatsHap; 3.2–5.1-fold versus HapCUT2). LongPhase 2 also had 1.3–1.8-fold lower block-wise Hamming distances than these tools at every coverage (Fig. 2b). The separation in switch error rates was largest in regions newly covered by the v5.0q benchmark (Fig. 3d,e).

Lower switch error rates were accompanied by similar block contiguity (Fig. 2c). Block N50 increased from 0.84–0.95 Mb at 10 $\times$ to 2.72–2.86 Mb at 60 $\times$ across the three tools. The blocks of LongPhase 2 were up to 11% shorter than those of WhatsHap at 10–20 $\times$ and 1–5% longer at 30–60 $\times$. The low-coverage shortfall arose from block splitting during GNN correction. Before correction, LongPhase 2 already had 1.9–2.5-fold lower switch error rates than WhatsHap, with block N50 between 3% shorter and 7% longer at every coverage (Supplementary Table 8). Co-phasing indels, SVs and 5mC with SNVs extended the blocks of LongPhase 2 to 6–50% beyond those of WhatsHap in SNV-only mode, while retaining 2.5–3.1-fold lower SNV switch error rates (Supplementary Tables 8 and 10). We assess the accompanying differences in phased fraction below (Fig. 4).

The reduction relative to WhatsHap and HapCUT2 persisted on chr17, chr21 and chr22, which were excluded from GNN training. All tools had higher switch error rates on these chromosomes than genome-wide. At 10–20 $\times$, LongPhase 2 had 2.8–3.9-fold lower rates than WhatsHap and 3.3–4.9-fold lower rates than HapCUT2. At 30–60 $\times$, rates were 2.7–4.9-fold lower than those of these comparators, although each coverage had only one run and 37–83 switch errors by LongPhase 2 (Supplementary Table 15).

Ralphi had higher switch error rates at 10–20 $\times$, whereas Margin had lower rates at low coverage with shorter blocks (Supplementary Fig. 9). Ralphi²⁷ had 2.5–3.5-fold higher switch error rates than LongPhase 2. Margin⁹ had the lowest rate at 10 $\times$ (0.069% versus 0.082%) and matched LongPhase 2 at 12 $\times$, but its rate decreased little thereafter and was 1.2–1.5-fold higher at 14–20 $\times$. Its blocks were 2.1–3.2-fold shorter throughout. On the held-out chromosomes, Margin also had lower rates at 10–12 $\times$ (0.099% versus 0.127% at 10 $\times$), with 3.3-fold shorter blocks (Supplementary Table 15). GCphase was excluded because it treats every input record as a heterozygous SNV (Methods).

LongPhase 2 also required less time at the coverages with available timing data (Supple-

mentary Fig. 10 and Supplementary Table 16). With 24 threads, SNV phasing and GNN correction took 4.6–17.6 min across 10–60 $\times$, compared with 25.2–113.8 min for WhatsHap and 24.1–96.4 min for HapCUT2 on one thread. CPU time, which accounts for computation across threads, was 1.8–4.5-fold lower than for WhatsHap and 1.8–2.8-fold lower than for HapCUT2. Between 10 and 60 $\times$, CPU time increased 2.5-fold for LongPhase 2, compared with 4.5-fold for WhatsHap and 4.0-fold for HapCUT2. On one thread, LongPhase 2 took 8.1 min at 10 $\times$ and 18.0 min at 60 $\times$, respectively 3.0–3.1-fold and 5.4–6.3-fold less than the other tools. Parallel execution increased memory use: peak memory was 36.0–63.0 GiB with 24 threads, compared with 6.2 and 7.4 GiB in the two single-thread runs, 1.5–4.0 GiB for WhatsHap and 6.4–19.4 GiB for HapCUT2.

The reduction in indel-pair switch error rates relative to WhatsHap increased with coverage. Adding indels lengthened the blocks of WhatsHap by 31–59% but increased its SNV switch error rate by 17–20% at every coverage (Fig. 2c,f). With the same indels, LongPhase 2 had 3.0–3.7-fold lower SNV switch error rates (Fig. 2d, solid lines). The difference was larger for consecutive variant pairs containing an indel: rates decreased from 0.101% to 0.030% for LongPhase 2, but remained at 0.75–0.91% for WhatsHap (dotted lines). The reduction relative to WhatsHap therefore increased from 8-fold at 10 $\times$ to 31-fold at 60 $\times$ (Supplementary Table 14). Indel Hamming distance also decreased for LongPhase 2, from 6.8% to 3.5%, but remained at 9.0–13.8% for WhatsHap (Fig. 2e).

With SNVs and indels, WhatsHap produced longer blocks than LongPhase 2, but the difference narrowed with coverage: its blocks were 28–34% longer at 10–18 $\times$ and 14% longer at 60 $\times$ (Fig. 2f). Adding SVs and 5mC, which WhatsHap cannot use, brought the blocks of LongPhase 2 closer to those of WhatsHap with indels: they were 2–21% shorter across coverages and 2–3% shorter at 50–60 $\times$. SNV switch error rates remained 3.0–3.7-fold lower (0.027% versus 0.097% at 60 $\times$; dashed line).

Joint SNV and 5mC phasing by LongPhase 2 had 1.5–1.7-fold lower switch error rates than MethPhaser at every coverage (Fig. 2g). MethPhaser²¹ adds methylation evidence to existing SNV phase blocks; we supplied the SNV-only output of LongPhase 2 before GNN correction (Methods). The Hamming distance of LongPhase 2 was 1.4-fold lower at 10 $\times$ and 2.1–3.0-fold lower at 20–60 $\times$ (Fig. 2h). MethPhaser produced longer blocks, with N50 1.1-fold that of LongPhase 2 at 10 $\times$ and 1.3–1.8-fold at 20–60 $\times$ (0.96–5.49 versus 0.88–3.09 Mb; Fig. 2i and Supplementary Fig. 11d). Phased fractions differed by approximately 0.2 percentage points or less (Supplementary Fig. 11b).

Most of the switch-error difference arose from GNN correction. MethPhaser joined input blocks without changing phase within them. These joins increased switch errors by 2–15% and Hamming distance by 4% at 10 $\times$ and 1.8–2.8-fold at 20–60 $\times$. Before GNN correction, joint phasing and the SNV-only input to MethPhaser differed in switch-error count by at most 2% (Supplementary Fig. 11). GNN correction accounted for 63–95% of the switch-error difference between the final outputs, with MethPhaser’s joins accounting for most of the remainder.

Without retraining the nanopore-derived network, LongPhase 2 had 2.2–2.6-fold lower SNV switch error rates than WhatsHap on HiFi data (Fig. 2j and Supplementary Table 11). Rates decreased from 0.052% to 0.034% for LongPhase 2 and from 0.128% to 0.077% for WhatsHap

between 10 and 50 $\times$. The difference narrowed from 2.5–2.6-fold at 10–30 $\times$ to 2.2–2.3-fold at 40–50 $\times$. Hamming distances were 0.7–1.2% for both tools; LongPhase 2 had lower values at 10–40 $\times$ and a higher value at 50 $\times$ (Fig. 2k). Block N50 increased from 340 to 580 kb and differed by at most 3% between tools (Fig. 2l). HiFi blocks were 2.5–4.3-fold shorter than nanopore blocks, consistent with the shorter reads.

Benchmark choice changes the measured separation between phasers

Against GIAB v4.2.1, the three phasers converged as coverage increased (Fig. 3a). LongPhase 2 made 1,635–1,820 switch errors across coverages, whereas WhatsHap and HapCUT2 decreased from 2,773 and 3,050 at 10 $\times$ to 1,801 and 2,088 at 60 $\times$. The ratio between the highest and lowest switch error rates fell from 1.7 to 1.2–1.3 at 30–60 $\times$. Against v5.0q, errors decreased for all tools, including from 1,518 to 496 for LongPhase 2, and the rate ratio increased from 3.3 to 4.3–4.8 (Fig. 3b). Relative to v4.2.1, v5.0q reduced errors for LongPhase 2 but increased them for HapCUT2 at every coverage and for WhatsHap below 60 $\times$. Hamming distance changed little (Supplementary Table 9).

Coincident switch errors accounted for much of the convergence under v4.2.1. The fraction of LongPhase 2 switch errors shared by both other tools increased from 63% at 10 $\times$ to 93% at 60 $\times$. At 60 $\times$, these errors accounted for 86% of WhatsHap errors and 75% of HapCUT2 errors; corresponding shares under v5.0q were only 37%, 10% and 8% (Fig. 3c). In shared benchmark regions at 30–60 $\times$, each tool made 1,127–1,492 switch errors against v4.2.1 but 32–393 against v5.0q (Supplementary Table 13). The truth sets thus differ in phase at many shared sites, although possible v4.2.1 phase-transfer errors have not been checked in the reads.

The separation between phasers also depended on genomic context. Within the v5.0q benchmark regions, LongPhase 2 had the lowest switch error rate of the three tools at every coverage. Rates were 2.1-fold lower than for WhatsHap and 2.6-fold lower than for HapCUT2 at 10 $\times$, increasing to 3.5–4.0-fold and 8.0–10-fold lower at 30–60 $\times$ (Supplementary Table 13). Among GIAB genomic strata²⁵, the separation was largest in segmental duplications (4.3–6.5-fold at 10 and 60 $\times$) and smallest outside difficult regions (1.7–2.0-fold; Fig. 3d,e). In regions newly covered by v5.0q³², WhatsHap and HapCUT2 had 5.6–9.3 and 7.5–12 times the switch error rate of LongPhase 2, compared with 1.7–2.6 and 2.1–9.8 times in shared benchmark regions. These comparisons involved 15–41 switch errors for LongPhase 2 in the newly covered regions per coverage.

Outside the benchmark regions of both truth sets, where the truth phase is least certain, switch errors accounted for an increasing share of each tool’s total as coverage rose. This share increased from 29% to 70% for LongPhase 2 and from 43–46% to 62–75% for the other tools between 10 and 60 $\times$. Rates in these regions were 0.8–1.2% for LongPhase 2 and 2.6–4.9% for the other tools. These sites contributed 50–56% of the genome-wide difference in switch-error counts at 10 $\times$ and 60–77% at 60 $\times$. We therefore treat this component of the comparison as exploratory (Supplementary Table 13).

LongPhase 2 phases slightly fewer benchmark SNVs and places fewer genotype errors on haplotypes than WhatsHap

LongPhase 2 phased slightly fewer benchmark heterozygous SNVs than WhatsHap and HapCUT2 despite identical input calls. We measured completeness as the fraction of benchmark heterozygous SNVs phased. WhatsHap and HapCUT2 phased 0.7–1.0 percentage points more than LongPhase 2 at every nanopore coverage. For all three tools, the phased fraction rose steeply up to $16\times$ and plateaued at 91–92% from $20\times$ (Fig. 4a and Supplementary Table 8). Most of the difference preceded GNN correction (0.6–0.9 percentage points); correction added 0.1–0.3. On HiFi data, WhatsHap phased 0.4–0.5 percentage points more, and both tools increased from 87% at $10\times$ to 92% at $20\text{--}50\times$ (Fig. 4b). With indels, WhatsHap phased 2.4–4.5 percentage points more benchmark heterozygous indels, with a larger difference at higher coverage (Supplementary Table 9).

A larger share of the calls phased by LongPhase 2 matched benchmark heterozygous SNVs. Among heterozygous SNV calls phased on chr1–22, the share matching a benchmark heterozygous SNV increased from 89.4% to 93.1% for LongPhase 2 and from 86.2% to 91.5% for WhatsHap (Fig. 4c). The share was higher for LongPhase 2 in every replicate, although the difference narrowed from 3.2 percentage points at $10\times$ to 1.5 at $50\text{--}60\times$. F1, the harmonic mean of this share and completeness, was also higher in every replicate. The difference decreased from 0.8–1.0 percentage points at $10\text{--}18\times$ to 0.3 at $50\text{--}60\times$ (83.2–92.2% for LongPhase 2 and 82.3–91.9% for WhatsHap; Supplementary Table 8).

LongPhase 2 also phased fewer genotype errors than WhatsHap. These errors place an incorrect allele on a haplotype when a homozygous site is called and phased as heterozygous³². At positions with a homozygous v5.0q genotype on chr1–22, WhatsHap phased 27,135, 13,706 and 11,815 heterozygous calls at 10, 30 and $60\times$, compared with 14,444, 9,828 and 8,497 for LongPhase 2 (1.4–1.9 times as many; Fig. 4d). WhatsHap also phased 1.3–1.4 times as many calls with a different allele in v5.0q or no v5.0q record. The latter category included 190,344 calls for WhatsHap and 152,798 for LongPhase 2 at $60\times$; benchmark absence alone does not establish a genotype error.

Calls phased by only one tool explained these differences (Supplementary Table 17). WhatsHap phased 122,000–149,000 calls that LongPhase 2 left unphased; 15–20% matched benchmark heterozygous SNVs. Among benchmark-matching calls in blocks with at least two assessed variants, 22.9%, 15.9% and 10.0% were phased incorrectly at 10, 30 and $60\times$, three to six times its overall rate. Conversely, LongPhase 2 phased 38,000–54,000 calls that WhatsHap left unphased. About 90% were absent from the benchmark, and 92–93% lay in segmental duplications or outside one-to-one assembly alignments. Among the 991–1,565 benchmark-matching SNVs assessed in such blocks, 13.9–16.7% were phased incorrectly. Calls phased by either tool alone also overlapped that tool’s switch-error intervals 2.0–9.6 times more often than its phased calls overall.

At higher sequencing coverage, calls left unphased by LongPhase 2 alone included more high-depth sites than those left unphased by WhatsHap alone (Supplementary Fig. 12). At $10\times$, 64–71% of calls in each set had depth below $20\times$. At 30 and $60\times$, this remained true for 53–60% of calls left unphased by WhatsHap alone, compared with 12–16% of those left

unphased by LongPhase 2 alone. The latter set had a depth peak near the sequencing coverage and included more high-depth calls. At 60×, 29% had depth of at least 90×, and 15.6% reached the variant caller’s depth cap. Only 3% of calls left unphased by WhatsHap alone reached 90×.

The excess depth persisted when measured relative to sequencing coverage. At 30 and 60×, 28% and 17% of calls left unphased by LongPhase 2 alone had at least twice the sequencing coverage, compared with 7% and 2% for WhatsHap alone. Such depth can arise when reads from two paralogous copies map to one locus, a known source of false variant calls³². At 60×, 3,684 calls left unphased by LongPhase 2 alone had depth of at least 90× within v5.0q benchmark regions; 92.0% were absent from the benchmark VCF and thus classified as false positives by this benchmark. However, calls left unphased by LongPhase 2 alone overlapped segmental duplications or regions outside one-to-one assembly alignments less often than those left unphased by WhatsHap alone (54–75% versus 92–93%). The source of the excess depth remains unresolved.

At 60×, SNVs unphased by GNN correction lie mostly outside one-to-one assembly alignments

Most SNVs unphased by GNN correction were absent by position from the v5.0q VCF. GNN correction unphased 55,756–58,162 SNVs per genome at 10–60× (one replicate per coverage). This step accounted for 15–18% of the calls left unphased by LongPhase 2 alone at 10, 30 and 60×; the rest were withheld before correction. Among GNN-unphased SNVs, the share absent by position from the v5.0q VCF increased from 83.0% to 93.0% with coverage, while the heterozygous share decreased from 11.9% (6,650) to 4.0% (2,215); the remainder were homozygous (Fig. 4e). Allele matching reduced the heterozygous counts to 6,405 and 2,055. Benchmark absence alone does not establish a call error because the benchmark depends on assembly alignment and variant representation.

An exploratory analysis of T2T-HG002 assembly alignments placed 11.8% of the 56,076 SNVs unphased at 60× in regions where both haplotypes aligned one-to-one to GRCh38. This compared with 92.6% of a random sample phased before correction (odds ratio for lying outside these regions, 94; Fig. 4f and Supplementary Table 12). On autosomes alone, the shares were 9.6% and 92.9% (odds ratio, 124), with enrichment outside one-to-one alignments on every autosome (Fig. 4g). Sites outside the one-to-one alignment regions clustered in pericentromeric satellites, particularly at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 6,601 unphased SNVs within one-to-one alignments, 3,421 (52%) had no assembly variant at the same start coordinate.

This assembly comparison identifies genomic context but does not independently validate phase or allele identity. Coordinate matching can miss variants represented at another position. The background sample included sites later unphased by correction, which biases the comparison towards no difference. The one-to-one regions came from raw assembly alignments and retain assembly errors removed during benchmark curation. Moreover, v5.0q derives from the same assembly. Thus, the analysis extends the assessment beyond benchmark regions, but cannot establish whether the excluded phase assignments were correct.

GNN correction reduces genome-wide SNV switch error rates by one-third and limits indel-associated errors

To separate the contribution of GNN correction from initial phasing, we compared LongPhase 2 with and without this step. Before correction, SNV switch error rates were already 1.9–2.5-fold lower than for WhatsHap and 2.2–3.3-fold lower than for HapCUT2 (Supplementary Fig. 13b and Supplementary Table 8). Correction reduced rates by a further 31–33% at every coverage (30–36% across individual replicates at 10–20×). The associated loss of completeness decreased with coverage, from 0.26 percentage points at 10× to 0.09–0.11 at 30–60×. Block N50 decreased by 5–9% at 10–20× and 2–5% at 30–60× (Supplementary Fig. 13a,d).

The correction effect was smaller on held-out chromosomes and HiFi data. On held-out chromosomes, correction removed 20–23% of switch errors at 10–20× and 14–20% at 30–60× (9–15 per single run). It withheld 10.5 phased SNVs per removed error, compared with 8.3 genome-wide (Supplementary Table 15). Every tool had higher switch error rates on these chromosomes than genome-wide. The smaller reduction may therefore reflect both chromosome difficulty and sequence absent from training. On HiFi data, correction removed 19–25% of switch errors while reducing completeness by at most 0.1 percentage points and block N50 by 0.6–1.6% (Supplementary Fig. 13e,f).

Indels contributed most to both contiguity and error across seven combinations of variant classes, and GNN correction reduced the additional errors (Supplementary Figs. 14 and 15 and Supplementary Table 10). Before correction, adding SVs increased block N50 by 1–2% and adding 5mC by 2–13%, with SNV switch error rates changing by at most 3%. Indels increased N50 by 29–51% (32–75% with SVs and 5mC), but increased switch error rates by 24–34%. GNN correction reduced rates by 30–33% without indels and 34–42% with indels, removing 41–71% of the additional error and 24–39% of switch errors at indel pairs (Supplementary Table 14). Block N50 decreased by 12–22% with indels and 1–9% without them as unphasing disconnected blocks.

After correction, each evidence class still increased contiguity (Supplementary Fig. 15e–h and Supplementary Table 10). Adding 5mC kept SNV switch error rates within 4% of SNV-only phasing and increased N50 by 2% at 10× and 8–14% from 18×. Indels increased N50 by 15–32% and switch error rates by 12–21%. Four-class phasing produced the longest blocks at every coverage, 19–47% longer than SNV-only blocks (4.10 versus 2.86 Mb at 60×), with 14–27% higher switch error rates. Adding SVs and 5mC to indels changed indel-pair switch error rates by at most 6% (Supplementary Table 14). All configurations retained at least 2.5-fold lower SNV switch error rates than WhatsHap with or without indels.

The phased fraction varied by variant class in four-class phasing. Before GNN correction, LongPhase 2 phased 30.8–45.5% of benchmark heterozygous indels, 99.0–99.4% of the heterozygous 5mC sites called by `modcall` and 68.1–70.1% of the heterozygous SVs called by `Sniffles2`; GNN correction then unphased 1.4–4.6% of the phased 5mC sites and 2.4–3.8% of the phased SVs[Add the phased SV and 5mC counts per configuration and coverage (audit table `q10_sv_mod.tsv`) to Supplementary Data 1].

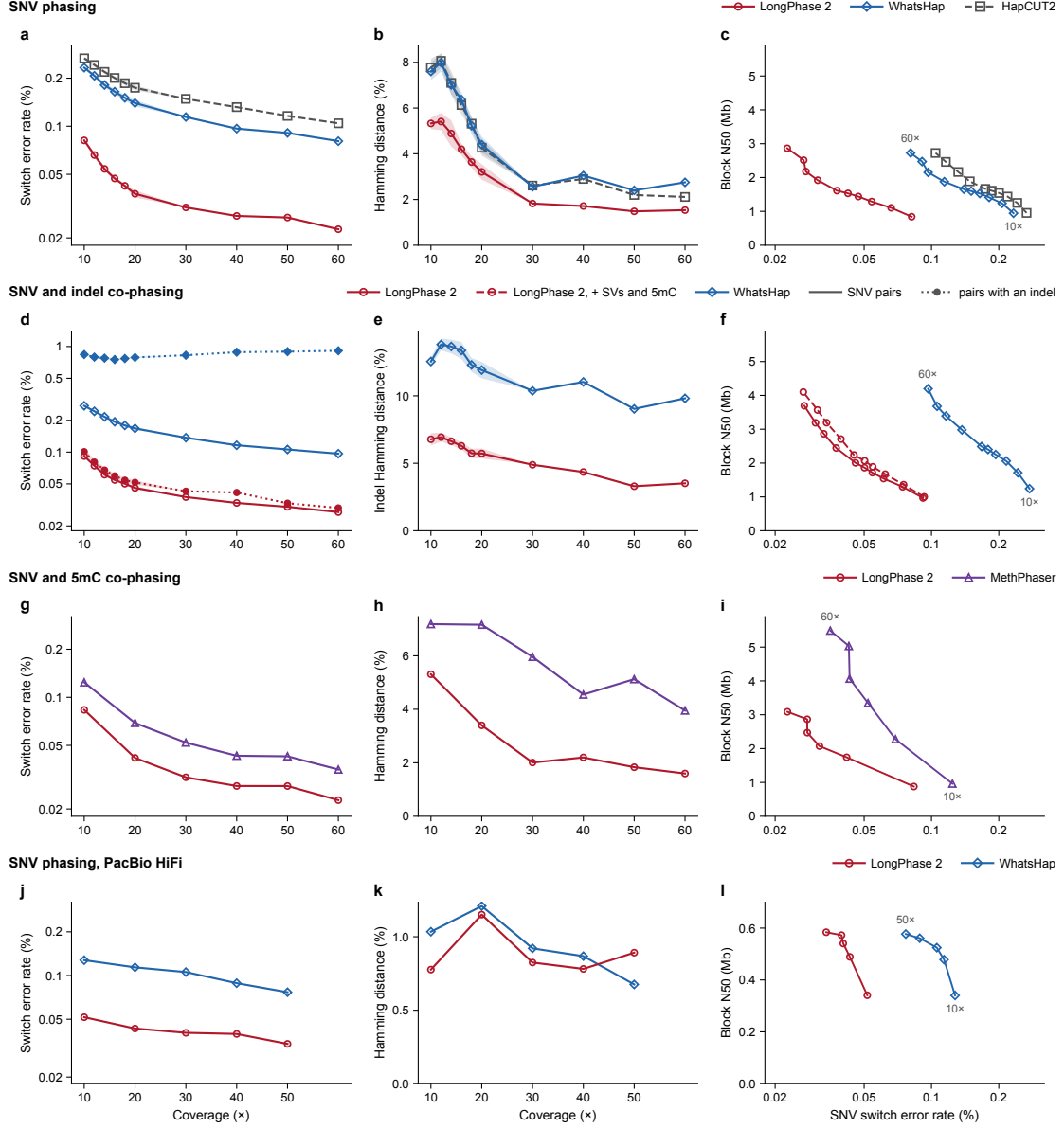


Figure 2: Accuracy and contiguity of LongPhase 2 compared with other phasers for each type of phasing evidence and on PacBio HiFi data. HG002 nanopore R10.4.1 reads down-sampled to 10–60× (**a–i**) and PacBio HiFi reads down-sampled to 10–50× (**j–l**); identical variant calls for every tool, scored against the T2T-HG002-derived v5.0q benchmark. Nanopore values at 10–20× are means of ten down-sampling replicates with the s.d. as a light band; all other values come from one down-sampling (Methods). Columns show the switch error rate (log scale), block-wise Hamming distance, and SNV switch error rate against phase-block N50, one point per coverage from the lowest (bottom right) to the highest (top left). Phased fractions are shown in Fig. 4. **a–c**, SNV phasing compared with WhatsHap and HapCUT2. **d–f**, SNV and indel co-phasing compared with WhatsHap; in **f**, the dashed line adds SVs and 5mC, which WhatsHap cannot use, to the co-phasing of LongPhase 2. In **d**, solid lines score consecutive SNV pairs and dotted lines consecutive pairs that include an indel; **e** refers to indels (Supplementary Table 14). **g–i**, SNV and 5mC co-phasing compared with MethPhaser, replicate 1; MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction (Methods). **j–l**, SNV phasing of PacBio HiFi (Revio) data compared with WhatsHap, with variants called at each coverage; LongPhase 2 uses the network trained on nanopore data. The N50 axis of **l** differs from those of **c**, **f** and **i**.

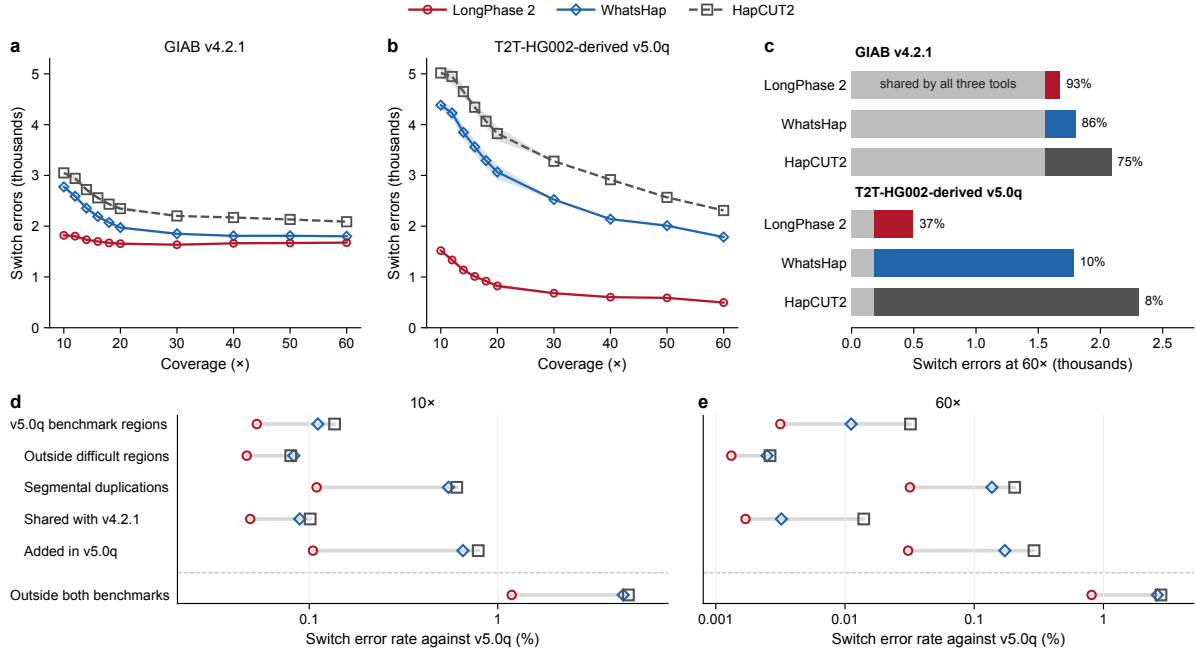


Figure 3: The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark. SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2; both truth VCFs used in full. **a,b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**). **c**, Switch errors at 60x; grey, switch errors made by all three tools at the same SNV pair, with their share of each tool's total. **d,e**, Switch error rate against v5.0q at 10x (**d**) and 60x (**e**) inside the v5.0q benchmark regions and in subsets of them: outside difficult regions and in segmental duplications (GIAB v3.6 strata), and in the parts that v4.2.1 also covers or that v5.0q adds. Below the dashed line, outside the benchmark regions of both truth sets, where the truth phase is least certain (exploratory). A variant pair is assigned to a region when both of its variants lie in it (Supplementary Table 13).

Calls phased by LongPhase 2 and WhatsHap

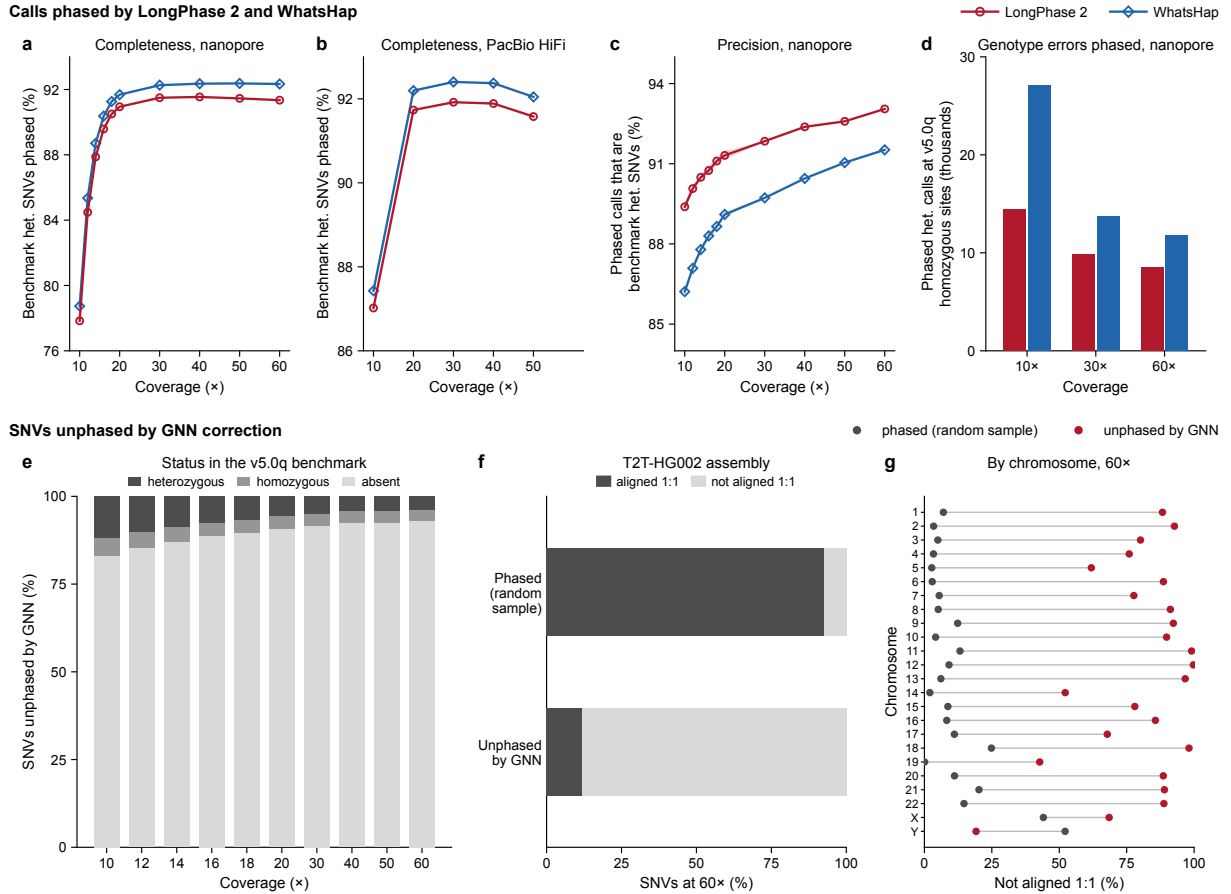


Figure 4: Heterozygous SNV calls phased by LongPhase 2 and WhatsHap, and SNVs unphased by GNN correction. SNV-only phasing of HG002 nanopore R10.4.1 data (**a**, **c–g**) and PacBio HiFi data (**b**). **a–c**, coverages and replicates as in Fig. 2; **d–g**, replicate 1. **a,b**, Fraction of the heterozygous SNVs of the v5.0q benchmark phased from nanopore (**a**) and HiFi (**b**) data; both tools received the same heterozygous calls. **c**, Share of the phased calls on chr1–22 that match a benchmark heterozygous SNV in position and alleles. **d**, Phased heterozygous calls on chr1–22 at positions where v5.0q has a homozygous genotype (genotype errors; Supplementary Table 17). **e**, SNVs phased before and unphased after GNN correction, by their genotype in the v5.0q benchmark VCF, matched by position without a benchmark BED. **f**, At 60x, these SNVs and a random sample of SNVs phased before correction, inside or outside the regions where both haplotypes of the T2T-HG002 assembly align one-to-one to GRCh38 (exploratory). **g**, The share outside these regions for each chromosome (Supplementary Table 12).

Discussion

LongPhase 2 combines genetic variation and allele-specific methylation in a shared phasing graph with selective withholding of phase assignments predicted to be unreliable. This approach extends methods that use methylation or SVs to connect blocks in regions with few heterozygous SNVs^{2,11,20,21,33}. Whereas LongHap adds methylation after sequence phasing²³, LongPhase 2 integrates both during haplotype construction. The evidence classes contributed differently: methylation improved contiguity at similar SNV switch error rates, whereas indels produced most of the gains in contiguity and increases in error. The same trade-off in WhatsHap suggests that difficulties in indel representation or modelling may affect both methods. In joint SNV and indel phasing, LongPhase 2 with GNN correction had 8–31-fold lower switch error rates than WhatsHap at variant pairs containing an indel. GNN correction reduced the additional error associated with indels, and improved indel calling could further improve the balance^{15,34}.

Selective withholding requires a distinction between evidence agreement and confidence in phase. LongPhase 2 reports phasing entropy and weighted vote totals. Unlike model-based confidence estimates from HapCUT2⁸ and statistical phasers⁵, low entropy indicates agreement among available votes, not necessarily strong evidence or a low error probability. The GNN uses neighbouring graph and sequence features and may detect candidates missed by an entropy threshold alone. However, its added value over simpler withholding rules and HapCUT2’s confidence pruning remains untested at matched variant retention. Its onset-prediction score is neither calibrated nor included in the output. These distinctions limit the use of the reported scores as quantitative measures of error risk.

Withholding variants and splitting blocks also change which variant pairs can be assessed. Switch error rates must therefore be considered together with completeness and contiguity. Before GNN correction, LongPhase 2 already had 1.9–2.5-fold lower switch error rates than WhatsHap at similar block N50. Most additional SNV calls phased by WhatsHap did not match a benchmark heterozygous SNV. Among assessable benchmark heterozygous SNVs phased by WhatsHap alone, 10–23% were phased incorrectly, three to six times its overall rate. LongPhase 2 had a similar error rate among the corresponding SNVs phased by it alone, but this set was 13–25 times smaller. Margin’s lower error rate but shorter blocks at low coverage illustrates a different balance. At 60×, most SNVs unphased by the GNN lay outside one-to-one assembly alignments, where our analysis cannot establish whether their phase was correct.

Benchmark choice changed the apparent separation between methods. At 30–60×, the highest and lowest switch error rates differed by 1.2–1.3-fold against GIAB v4.2.1 but by 4.3–4.8-fold against v5.0q. Many apparent errors under v4.2.1 occurred at the same variant pairs across tools, and most disappeared when the same outputs were scored against v5.0q in shared benchmark regions. All three phasers agreed with v5.0q at most of these sites, including WhatsHap and HapCUT2, which have no parameters trained on either truth set. Errors in v4.2.1 phase transfer are one possible explanation, but shared reads, alignments and variant calls could also contribute to correlated outputs. Independent checks with reads or parental data are needed to distinguish these explanations.

Within v5.0q benchmark regions, LongPhase 2 had 2.1- to 10-fold lower switch error rates than WhatsHap and HapCUT2, with the largest separation in newly covered regions. How-

ever, much of the genome-wide difference arose outside the benchmark regions of both truth sets, where the phase is least certain. This component remains exploratory. The v5.0q release evaluated variant calls rather than phase³². Our results support comparing phasers against assembly-phased truth sets, with phase errors reported separately inside and outside benchmark regions and stratified by genomic context²⁵. Reporting which calls receive phase assignments also captures genotype errors propagated onto haplotypes, which phase-error rates alone do not measure.

Generalization remains the main evaluation limit. All data came from HG002. The network was trained on nanopore runs from the same down-sampling series, using labels from the evaluation benchmark. Genome-wide metrics include 17 training chromosomes. On held-out chromosomes, the advantage over WhatsHap and HapCUT2 persisted, but correction removed fewer errors at a higher cost in withheld SNVs. These chromosomes were excluded from GNN training but contributed to earlier tuning of the graph constants. The HiFi results without retraining support transfer between platforms within HG002, but do not establish transfer to other individuals. Evaluation in other individuals against an independent benchmark, with fixed weights and thresholds, is needed. Subsampling variability was not estimated above 20× for nanopore data or at any HiFi coverage.

Further work should test both the scope of co-phasing and its downstream effects. SV and 5mC evaluation measured phased fractions rather than phasing accuracy, and neither class had supervised training labels. Methylation co-phasing was assessed only on autosomes of a male lymphoblastoid cell line²¹; the method assumes diploidy and balanced allele fractions. Compatibility with pipelines that use LongPhase^{14,16,18,19} enables tests of selective withholding in haplotype-aware somatic calling. More accurate germline phase could improve tumour-read assignment, but withholding phase may reduce information available to the caller. Varying the germline phaser while fixing the somatic caller would quantify this trade-off. A calibrated per-variant confidence measure could further allow downstream methods to account explicitly for uncertainty in inherited phase.

Methods

Overview of LongPhase 2

LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs, small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations are represented together in one weighted graph per chromosome. Initial haplotypes are resolved from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a graph neural network (GNN) then scores candidate phase-disagreement onsets. Selected variants are unphased, never flipped; if this disconnects a haplotype block, the surviving components receive updated phase-set identifiers. The detailed module rules are given in Supplementary Methods 1–6. The method extends the graph formulation of LongPhase 1¹¹.

Input variant calls

Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant⁹; homozygous and filtered records were carried through unchanged but did not enter the graph. Indels in short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat indel can link the wrong alleles. SVs were taken from Sniffles2¹² or cuteSV¹³ with supporting read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded. The exact eligibility and repeat annotation rules are in Supplementary Method 2.

Calling allele-specific methylation

Modification probabilities were read from the MM/ML tags produced by Dorado³⁵ and carried through alignment. A CpG was called methylated when its 5mC probability was at least 0.8, unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand observations were mapped to one CpG coordinate. A site was retained as a candidate allele-specific site when methylated and unmethylated reads were sufficiently balanced and the ambiguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \quad (1)$$

where M_R , U_R and A_R are the three read counts. Candidates were then required to co-segregate with a neighbouring heterozygous marker. For a pair of sites, writing RR , RA , AR and AA for the four observed state combinations, linkage required

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9. \quad (2)$$

The same criterion was used for SNV-anchored and methylation-only linking. Only CpG dinucleotides were considered. Weak candidates were admitted in a later expansion pass, and consecutive CpGs were merged into one locus so that both strands shared one read list. The resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event or the caller's read list; and 5mC alleles from the methylation read lists. Supplementary alignments were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read's mismatch load R_m counted the heterozygous sites in the

interval at which it carried the alternate allele, and each variant was scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v . Sites with $\text{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v) , the four allele-pair supports were accumulated as w_{rr} , w_{ra} , w_{ar} and w_{aa} . The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV-5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at 10–60× and were not re-tuned for the reported experiments; the parameter tuning is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass). Reads were retained only with a majority above 0.65 and total evidence above 1.

For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}. \quad (4)$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

Phasing entropy was used to select candidate windows for refinement. Within each window, the GNN combined phasing evidence, genomic context and graph structure to identify phase-disagreement onsets for selective unphasing, as defined by the training labels. Positive targets include isolated disagreements and the first positions of consecutive disagreement runs. The resulting onset scores are not calibrated probabilities.

The GNN operated on merged SNV/indel, SV and 5mC records. A window of at most 41 variants (82 allele nodes) was opened around every phased variant with $\text{PE} \geq 0.8$, extending 20 variants on either side.

Each allele node carried 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2 , and six transforms of them); genomic context, derived from the reference sequence and input variant positions (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator); and four one-hot variant-class indicators. Six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, symmetrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 1–2.

The network used four GPS graph-transformer layers³⁶, each combining a local four-head GATv2 branch^{37,38} and a global four-head self-attention branch³⁹, followed by residual normalization and a 128–256–128 feed-forward block. The local GATv2 branch aggregates information along edges of the phasing graph, using edge features to weight the contributions of neighbouring allele nodes. The global self-attention branch allows each node to access information from all

nodes within the same window, including those without a direct graph connection. Together, these branches allow the model to combine the allele-linkage evidence represented by the graph with patterns of phasing support and genomic context across the window when predicting phase-disagreement onsets. A two-layer classifier produced the two-class softmax output. Exhaustive architecture and inference details are in Supplementary Method 4 and Table 3.

Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Because the zone is defined on genomic distance and not on variant index, the number of variants it holds depends on the local spacing. The two allele-node scores were averaged, followed by averaging across the windows in which the variant lay inside the scoring zone.

A variant with mean score at least 0.30 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, and SV and 5mC correction relies on the model trained on those labelled small variants. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model was trained and applied in binary mode, in which flip, unphase and break are merged into a single error class, matching the single action the module is allowed to take. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were sampled independently in each minibatch at a ratio of at most four per error node, and the resulting cross-entropy used focal weighting⁴⁰ with $\gamma = 1.5$ and label smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; the optimizer, stopping rule and environment are listed in Supplementary Table 4.

We used a normalized net-gain criterion for checkpoint selection, assigning unit benefit to each true-positive detection of a label-defined correction target and unit cost to each false-positive prediction. Checkpoints were selected on chr15–16 by maximizing

$$S = R \left(2 - \frac{1}{P} \right) \quad (5)$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are precision and recall for the label-defined error class. This expression applies for $P > 0.5$; otherwise the score was $P - 1$. The criterion serves as a proxy for correction utility (Supplementary Method 4).

Checkpoint details and classification metrics for validation and held-out chromosomes are reported in Supplementary Table 4.

The deployment threshold $B = 0.30$ was chosen on the 60 HG002 nanopore runs at 10–20× as the threshold that removed the most switch errors while withholding at most ten phased SNVs per removed error. It removed 33.1% of switch errors at 8.3 withheld SNVs per removed error over the whole genome, and 21.2% at 10.5, slightly above the limit, on the held-out chromosomes (Supplementary Method 4 and Supplementary Table 15).

Inference was implemented in C++ without external runtime dependencies; the export procedure and numerical agreement checks are described in Supplementary Method 4.

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) Oxford Nanopore R10.4.1 reads from the ONT `giab_lsk114_2022.12` release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado `dna_r10.4.1_e8.2_400bps_sup@v4.0.0` and 5mCG `5hmCG@v2`; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set by Oxford Nanopore with `minimap2 v2.24 (-y -ax map-ont)`, retaining reads with mean base quality ≥ 10 . The 60× data set is a random 91.14% subset of the 70.4× alignment (`samtools v1.15, view -s 0.9114`), and the 10–50× data sets were down-sampled from it with `samtools v1.23.141 view -s seed.fraction`. Ten replicates were used at 10–20× and one at 30–60×; variants were called separately at every coverage and replicate. SNVs and indels came from PEPPER-Margin-DeepVariant r0.8, which reports unphased genotypes, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from Long-Phase 2. The seeds were 1–9 and 867 at 10× (fraction 0.1666), 12–21 at 12× (0.200), 22–31 at 14× (0.233), 32–41 at 16× (0.266), 42–51 at 18× (0.300), 1–10 at 20× (0.3333), and 42 at 30, 40 and 50× (0.5, 0.6667 and 0.8333). PEPPER-Margin-DeepVariant used the `-ont_r10_q20` model for nanopore data and `-hifi` for HiFi data. HG002 PacBio HiFi (Revio) reads were taken from a 48× GIAB alignment (Supplementary Table 5), which was used as 50×; 10–40× data sets were down-sampled from it with `samtools v1.23.1 view -subsample` (seed 1; fractions 0.2083, 0.4167, 0.6250 and 0.8333), and variants were called separately at every coverage.

Accuracy was scored with `longphase compare` on autosomes 1–22 against two HG002 truth sets: GIAB v4.2.1²⁹, in the version with phase transferred from a hifiasm assembly (the v4.2.1 benchmark VCF itself is unphased), restricted to biallelic heterozygous records; and GIAB v5.0q³², derived from the diploid T2T-HG002 assembly³⁰, in which every heterozygous variant is phased. The truth VCFs were not restricted to their benchmark regions, and a call was scored

only where the truth VCF holds the same REF and ALT alleles in a heterozygous genotype. Both truth VCFs list heterozygous variants outside their benchmark regions (9.3% of the autosomal heterozygous records of v5.0q and 3.4% of v4.2.1), so the error counts include sites that the benchmark regions do not cover, among them satellite and segmental-duplication sequence that the T2T-HG002 assembly resolves³⁰. To locate switch errors and to score indels, every consecutive variant pair in an intersected block was recorded with a re-implementation of **longphase compare**, which reproduced its counts exactly in all 1,024 scored runs. A pair was assigned to a region when both variants lay in it; regions were the intersection and the differences of the two sets of benchmark regions, and the GIAB v3.6 genome stratifications²⁵. Indel switch errors were counted over consecutive pairs that included at least one indel, and the indel Hamming distance was computed in the block orientation chosen on all variants.

Comparison with other methods

WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0¹¹ and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.30, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 10, with and without correction. Indel co-phasing was also compared with WhatsHap’s default indel phasing. Margin v2.3.1⁹, Ralph²⁷ and MethPhaser²¹ were evaluated as described in Supplementary Table 5. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction instead of the WhatsHap or HapCUT2 phasing its authors recommend as input, so that MethPhaser and the joint SNV and 5mC phasing of LongPhase 2 started from the same SNV phasing. Without GNN correction, differences between them reflect how methylation is used, after phasing to bridge existing blocks (MethPhaser) or during phasing as additional graph evidence (LongPhase 2); with GNN correction, they also include its effect. GCphase⁴² was not included: it treats every input record as a heterozygous SNV and ignores the FILTER field, so with these call sets it phased about 1.2 million rejected reference calls per 10× genome as heterozygous variants, and a fair comparison would require a different input. The HiFi call sets were phased SNV-only with LongPhase 2 (**-pb**), with and without correction, and with WhatsHap; the GNN weights were trained on nanopore data. WhatsHap was run as **whatshap phase -ignore-read-groups -reference** with **-only-snvs** for SNV comparisons, and HapCUT2 as **extractHAIRS -ont 1** followed by **HAPCUT2** with default parameters, including its default likelihood-based pruning, on the PASS heterozygous SNVs. LongPhase 1.0 was re-run on every down-sampling replicate. All tools ran on the same workstation.

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Hamming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-

consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files, so it is not comparable between configurations with and without indels, which we therefore compared by SNV switch error rate. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6.

Analysis of unphased SNVs

An SNV was counted as unphased when it was phased in the pre-correction output and unphased after GNN correction. Such sites were classified against the v5.0q benchmark VCF, matched by chromosome and position and without a benchmark BED, as absent, homozygous or heterozygous (Fig. 4e); the heterozygous count was also reported with matching REF and ALT alleles, the rule used by `compare`. The analysis used the 10–60× outputs of replicate 1 with the final GNN model. At 60×, they were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one; alleles were not compared. All chromosomes, including chrX and chrY, were analysed, and autosome-only values were derived from the per-chromosome counts. A random sample of 50,000 SNVs phased before correction, which included sites that GNN correction later unphased, provided the background; enrichment was summarized by chromosome and 1-Mb bins. To compare tools, the heterozygous SNV calls with single-base alleles and genotype 0/1, which were identical in the two output VCFs, were grouped by the tool that phased them: both tools, LongPhase 2 only, WhatsHap only or neither; calls phased by WhatsHap only were further split by whether LongPhase 2 had already left them unphased before GNN correction. Each call was classified against the v5.0q chr1–22 VCF, without a benchmark BED, as heterozygous with the same position and alleles, heterozygous with other alleles, homozygous or absent; calls on chrX and chrY were counted separately. The calls phased by a tool were those phased by both tools plus those phased by that tool only; phased calls at positions with a homozygous v5.0q record were counted as genotype errors (Fig. 4c). For every SNV-only run of both tools, the share of the phased heterozygous SNV calls on chr1–22 that matched a benchmark heterozygous SNV (precision) was computed from the `compare` count of matched calls, and the F1 score as the harmonic mean of precision and the phased fraction. For the matched calls phased by one tool only, the phase assigned by that tool was scored as in `compare`, and a call was counted as wrongly phased when its phase was opposite to the majority orientation of its block relative to the truth. A switch-error interval extends from the first SNV of a switch-error pair to the base before the second; the share of the calls phased by one tool only inside that tool’s intervals was compared with the share of all heterozygous SNV calls phased by that tool. Calls were also annotated by read depth, by the v5.0q benchmark regions, by GIAB v3.6 segmental duplications and by the one-to-one dipcall alignment (Supplementary Data 1). [\[Add the script behind Fig. 4e \(`analyze\_unphased.py`\) to the repository; the scripts for Fig. 4c,d are](#)

in notes/unphased.]

Runtime and memory

Wall-clock time, CPU time and peak resident memory were measured with `/usr/bin/time`; reported LongPhase 2 times included alignment reading, graph construction, phasing, GNN correction and VCF writing, and excluded alignment, variant calling and down-sampling. All runs used one workstation (AMD Ryzen 9 9950X3D, 16 cores and 32 threads, 249 GB of memory, Ubuntu 24.04, input and output on a hard-disk drive). LongPhase 2 was compiled with g++ 13.3 at `-O3`. Times were taken once per coverage, on replicate 1 at 10–20× and on the single data sets at 30–60×, from the runs that produced the phased VCFs: LongPhase 2 with 24 threads and WhatsHap and HapCUT2 on one thread. LongPhase 2 was also timed on one thread at 10 and 60× with no other jobs running. HapCUT2 times are not available at 30 and 40× because the logs of these runs were not retained. WhatsHap and HapCUT2 have no thread option; running them on each chromosome in parallel would shorten their wall-clock time but not their CPU time, so CPU time is reported next to wall-clock time (Supplementary Fig. 10 and Supplementary Table 16).

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20× were summarized by their mean and standard deviation, while 30–60× and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60× or for HiFi. Tools compared at each coverage and replicate used the same sampled reads, so fold differences between tools and the effect of GNN correction at 10–20× were also computed within each replicate (Supplementary Table 8). No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model weights and no sampling.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data analysed in this study are publicly available: the GRCh38 no-alt analysis set from NCBI (https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/); the GIAB HG002 v4.2.1 phase-transfer VCF (https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/SupplementaryFiles/) and the v5.0q benchmark (https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/v5.0q/) from the GIAB FTP site; and the HG002 nanopore R10.4.1 reads and

alignments from the ONT open-data archive ([s3://ont-open-data/giab_lsk114_2022.12/](https://ont-open-data/giab_lsk114_2022.12/)) under a CC BY-NC 4.0 licence. The truth files used are listed in Supplementary Table 5.

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>; the repository includes a demonstration data set and workflow. The reported results correspond to release v2.1 (commit [46ba47025b2704ab00cf54a35d3d4fead3eb148c](https://github.com/twolinin/longphase/commit/46ba47025b2704ab00cf54a35d3d4fead3eb148c)). The GNN training code, the trained model and the scripts used for the analyses and figures are available in the same repository. [\[Add the training, analysis and figure-generation scripts to the public repository under an OSI-approved licence, with development-machine paths parameterized, before submission.\]](#)

References

- [1] Eisfeldt, J., Ek, M., Nordenskjöld, M. & Lindstrand, A. Toward clinical long-read genome sequencing for rare diseases. *Nature Genetics* **57**, 1334–1343 (2025).
- [2] Akbari, V. *et al.* Parent-of-origin detection and chromosome-scale haplotyping using long-read DNA methylation sequencing and Strand-seq. *Cell Genomics* **3**, 100233 (2023).
- [3] Logsdon, G. A., Vollger, M. R. & Eichler, E. E. Long-read human genome sequencing and its applications. *Nature Reviews Genetics* **21**, 597–614 (2020).
- [4] Kolmogorov, M. *et al.* Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nature Methods* **20**, 1483–1492 (2023).
- [5] Hofmeister, R. J., Ribeiro, D. M., Rubinacci, S. & Delaneau, O. Accurate rare variant phasing of whole-genome and whole-exome sequencing data in the uk biobank. *Nature Genetics* **55**, 1243–1249 (2023).
- [6] Porubsky, D. *et al.* Fully phased human genome assembly without parental data using single-cell strand sequencing and long reads. *Nature Biotechnology* **39**, 302–308 (2021).
- [7] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).
- [8] Edge, P., Bafna, V. & Bansal, V. HapCUT2: robust and accurate haplotype assembly for diverse sequencing technologies. *Genome Research* **27**, 801–812 (2017).
- [9] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).
- [10] Martin, M. *et al.* WhatsHap: fast and accurate read-based phasing. *bioRxiv* (2016).
- [11] Lin, J.-H., Chen, L.-C., Yu, S.-C. & Huang, Y.-T. LongPhase: an ultra-fast chromosome-scale phasing algorithm for small and large variants. *Bioinformatics* **38**, 1816–1822 (2022).

- [12] Smolka, M. *et al.* Detection of mosaic and population-level structural variants with Sniffles2. *Nature Biotechnology* **42**, 1571–1580 (2024).
- [13] Jiang, T. *et al.* Long-read-based human genomic structural variation detection with cuteSV. *Genome Biology* **21** (2020).
- [14] Zheng, Z. *et al.* Symphonizing pileup and full-alignment for deep learning-based long-read variant calling. *Nature Computational Science* **2**, 797–803 (2022).
- [15] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [16] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).
- [17] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [18] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [19] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [20] Akbari, V. *et al.* Megabase-scale methylation phasing using nanopore long reads and NanoMethPhase. *Genome Biology* **22** (2021).
- [21] Fu, Y. *et al.* MethPhaser: methylation-based long-read haplotype phasing of human genomes. *Nature Communications* **15** (2024).
- [22] Chen, H., Nie, F., Luo, F. & Wang, J. HapBridge: a methylation-guided approach for correcting switch errors and bridging phased blocks in long-read phasing. *bioRxiv* (2025).
- [23] Pfennig, A. & Akey, J. M. Methylation-aware long-read phasing significantly improves genome-wide haplotype reconstruction. *bioRxiv* (2026).
- [24] Olson, N. D. *et al.* Variant calling and benchmarking in an era of complete human genome sequences. *Nature Reviews Genetics* **24**, 464–483 (2023).
- [25] Dwarshuis, N. *et al.* The GIAB genomic stratifications resource for human reference genomes. *Nature Communications* **15** (2024).
- [26] Xue, H., Rajan, V. & Lin, Y. Graph coloring via neural networks for haplotype assembly and viral quasispecies reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35 (2022).

- [27] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).
- [28] Zook, J. M. *et al.* An open resource for accurately benchmarking small variant and reference calls. *Nature Biotechnology* **37**, 561–566 (2019).
- [29] Wagner, J. *et al.* Benchmarking challenging small variants with linked and long reads. *Cell Genomics* **2**, 100128 (2022).
- [30] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).
- [31] Aganezov, S. *et al.* A complete reference genome improves analysis of human genetic variation. *Science* **376** (2022).
- [32] Olson, N. D. *et al.* The Genome in a Bottle HG002 assembly-based variant benchmark set enables comprehensive benchmarking of small and structural variants. *bioRxiv* (2026).
- [33] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [34] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).
- [35] Oxford Nanopore Technologies. Dorado: high-performance, easy-to-use, open source base-caller for Oxford Nanopore reads. <https://github.com/nanoporetech/dorado> (2025).
- [36] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).
- [37] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
- [38] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).
- [39] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
- [40] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).
- [41] Danecek, P. *et al.* Twelve years of SAMtools and BCFtools. *GigaScience* **10** (2021).
- [42] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).